[INCH-POUND]
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SUPERSEDING
MIL-D-4725C
4 February 1984

COMMERCIAL ITEM DESCRIPTION

DIMPLING MACHINE, HOT AND COLD PROCESS, PORTABLE

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-D-4725C for all federal agencies.

1. SCOPE. This commercial item description (CID) covers portable hot and cold process dimpling machines complete with tooling, accessories, and the controls necessary for automatic regulation of temperature, pressure, and dwell time for dimpling aircraft structural metal alloys.

2. CLASSIFICATION. The portable dimpling machines shall be of the following types. The type to be furnished shall be as specified (see 7.2(b)).

Type I - Squeeze gun
Type II - Impact gun
Type III - Blind-pull gun

3. SALIENT CHARACTERISTICS

3.1 General requirements. The dimpling machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The dimpling machine shall include, as a minimum, the following components:

3.2.1 Gun.

3.2.1.1 Squeeze gun. The type I dimpling machine shall include a squeeze gun with a throat depth not less than 2, 4, or 6 inches as specified (see 7.2(c)). The squeeze gun shall be capable of

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both cold and hot dimpling. The squeeze gun shall include switches, connectors, components, adjustments, and accessories necessary to form dimples in the materials shown in table I for the fastener types and sizes of die sets specified in 3.2.3.

3.2.1.2 Impact gun. The type II dimpling machine shall include an impact gun with a hand-held dolly or buck bar capable of holding self-aligning dies. The impact gun shall be capable of both cold and hot dimpling. The gun shall include switches, connectors, components, adjustments, and accessories necessary to form dimples in the materials shown in table II for the fastener types and sizes of die sets specified in 3.2.3.

3.2.1.3 Blind-pull gun. The type III dimpling machine shall include a blind-pull gun capable of cold and hot dimpling. The gun shall include switches, connectors, components, adjustments, and accessories necessary to form dimples in the materials shown in table II for the fastener types and sizes of die sets specified in 3.2.3.

3.2.2 Control unit. The dimpling machine shall include a portable control unit. The control unit shall include switches, indicators, regulators, timers, and other components necessary to preset, indicate, regulate, and automatically control die temperature, forming prepressure, dwell time, heating sequence, and final forming pressure.

3.2.2.1 Die heater control. The control unit shall include a die heater control that allows cold dimpling and hot dimpling with one or both die heaters in operation. Each die heater shall have an individual control that allows the die temperature to be selected manually in increments no greater than 25 degrees Fahrenheit between a range of 0 and 700 degrees Fahrenheit and the selected temperature to be controlled automatically within ±5 degrees Fahrenheit under all operating conditions. The control system shall provide a continuous visual indication of the die temperature and shall prevent gun operation if the die temperature is not within the selected temperature range.

3.2.2.2 Dwell time. The control unit shall have an adjustable automatic reset dwell timer that regulates the time period of the heating cycle during the prepressure stage of the dimpling process. Unless otherwise specified (see 7.2(d)), the control unit shall allow selection of the dwell time in 1/4-second increments over a range of 0 to 15 seconds.

3.2.2.3 Pressure regulator. An air pressure control and regulator shall be provided.

3.2.3 Die sets. As shown in table I for the type I dimpling machine and shown in table II for type II and III dimpling machines, die sets shall be provided for the fastener types and sizes as specified (see 7.2(e)). The die sets shall be made of carbide or machined from wear-resistant alloy steel that is heat-treated to impart necessary hardness and toughness. The die sets shall be suitable for both cold and hot dimpling, and their performance shall not be affected by the die heater.

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TABLE I. Fastener types and sizes with minimum and maximum metal thickness requirements for the type I dimpling machine.

Fastener type and size	Metal thickness, inch											
	Aluminum alloy		Magnesium alloy		Corrosion-resistant steel				Titanium			
					Annealed		Half-hard		Commercial pure		Ti-8Mn alloy	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
MS-20426 rivet												
3/32	0.020	0.064	0.020	0.051	0.020	0.050	0.020	0.050	0.016	0.050	0.020	0.050
1/8	0.020	0.072	0.020	0.064	0.020	0.063	0.020	0.050	0.016	0.063	0.020	0.050
5/32	0.025	0.072	0.025	0.072	0.025	0.063	0.025	0.036	0.020	0.063	0.020	0.036
3/16	0.025	0.072	0.025	0.081	0.025	0.050	0.025	0.032	0.025	0.050	0.025	0.032
1/4	0.032	0.072	0.032	0.091	0.032	0.043	0.032	0.043	0.032	0.040	0.032	0.040
MS 24694 screw												
#8	0.020	0.072	0.025	0.081	0.016	0.050	0.016	0.036	0.016	0.050	0.016	0.035
#10	0.020	0.072	0.025	0.072	0.025	0.050	0.020	0.025	0.020	0.032	0.020	0.025
1/4	0.032	0.072	0.032	0.081	0.025	0.064	0.025	0.064	0.020	0.050	0.025	0.043
Hi-shear rivet												
3/16	0.020	0.064	0.025	0.072	0.020	0.050	0.020	0.032	0.020	0.050	0.020	0.032
1/4	0.032	0.072	0.025	0.072	0.025	0.050	0.025	0.043	0.025	0.051	0.025	0.043
5/16	0.025	0.072	0.025	0.064	0.025	0.064	0.025	0.064	0.025	0.064	0.025	0.064
Drill lok scru												
#8	0.025	0.040	0.020	0.040	0.020	0.040	0.020	0.040				
#10	0.020	0.051	0.020	0.051	0.020	0.051	0.020	0.051				
Rivnuts												
#2	0.025	0.051	0.020	0.051	0.020	0.051	0.020	0.051	0.020	0.051	0.020	0.051
#6	0.032	0.040	0.020	0.040	0.020	0.040	0.020	0.040				
#8	0.025	0.051	0.020	0.051	0.020	0.051	0.020	0.051				
#10	0.025	0.064	0.025	0.064	0.025	0.064	0.025	0.064				
Airlock												
#5	0.025	0.051	0.020	0.051	0.020	0.051	0.020	0.051	0.020	0.051	0.020	0.051
#7	0.025	0.064	0.025	0.064	0.025	0.064	0.025	0.064	0.025	0.064	0.025	0.064
Comloc	0.020	0.040	0.020	0.040	0.020	0.040	0.020	0.040	0.020	0.040	0.020	0.040
FA-5 DZUS stud	0.025	0.051	0.020	0.051	0.020	0.051	0.020	0.051				
FA-6 1/2 DZUS stud	0.032	0.064	0.025	0.064	0.025	0.064	0.025	0.064				
FA-5 DZUS spring	0.020	0.051	0.020	0.051	0.020	0.051	0.020	0.051				
FA-6 1/2 DZUS spring	0.025	0.064	0.025	0.064	0.025	0.064	0.025	0.064				

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TABLE II. Fastener types and sizes with minimum and maximum metal thickness requirements for type II and III dimpling machines.

Fastener type and size	Metal thickness, inch											
	Aluminum alloy		Magnesium alloy		Corrosion-resistant steel				Titanium			
					Annealed		Half-hard		Commercial pure		Ti-8Mn alloy	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
MS-20426 rivet												
3/32	0.020	0.051	0.015	0.051						0.040		0.032
1/8	0.020	0.064	0.020	0.064						0.040		0.032
5/32	0.025	0.072	0.025	0.072						0.051		0.032
3/16	0.025	0.091	0.025	0.091						0.051		0.040
1/4	0.032	0.091	0.032	0.102						0.051		0.040
MS-24694 screw												
#8	0.025	0.064	0.025	0.064						0.032		0.032
#10	0.032	0.081	0.032	0.081						0.032		0.032
1/4	0.032	0.091	0.032	0.091						0.032		
Hi-shear rivet												
3/16	0.020	0.091	0.020	0.091						0.032		0.032
1/4	0.025	0.091	0.025	0.102						0.032		0.032
5/16	0.032	0.091	0.032	0.091						0.040		0.040
Cowl fastener												
#2 airlock	0.025	0.091	0.025	0.091						0.032		0.032
#5 airlock	0.032	0.091	0.032	0.091						0.032		0.032
#7 airlock	0.032	0.081	0.032	0.091						0.032		0.032
Comloc	0.020	0.040	0.020	0.040	0.020	0.040	0.020	0.040	0.020	0.040	0.020	0.040
FA-5 DZUS stud	0.025	0.051	0.020	0.051	0.020	0.051	0.020	0.051				
FA-6 1/2 DZUS stud	0.032	0.064	0.025	0.064	0.025	0.064	0.025	0.064				
FA-5 DZUS spring	0.020	0.051	0.020	0.051	0.020	0.051	0.020	0.051				
FA-6 1/2 DZUS spring	0.025	0.064	0.025	0.064	0.025	0.064	0.025	0.064				

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3.2.4 Die holders. The dimpling machine shall have die holders that maintain die alignment during the dimpling process and that allow dies to be changed easily. As shown in table I for the type I dimpling machine and shown in table II for type II and III dimpling machines, the die holders shall have a means of die adjustment for the metal thicknesses as specified (see 7.2(f)). When specified (see 7.2(g)), tools for die adjustment shall be provided.

3.2.5 Die heaters. The type I dimpling machine shall be equipped with die heaters for both male and female dies and shall have dual capability of heating either die independently or both dies simultaneously. Type II and III dimpling machines shall be equipped with die heaters for heating only male dies. The heating element shall surround or contact the die and maintain the control unit preset temperature. Die heaters shall operate on a 115-volt, single-phase, 60-hertz circuit and have sufficient wattage to meet the incremental and maximum temperature requirements specified herein. Die heaters shall be removable for repair or replacement. When specified (see 7.2 (h)), tools for the installation and removal of die heaters shall be provided.

3.2.6 Equipment cart. Unless otherwise specified (see 7.2 (i)), an equipment cart shall be provided with the dimpling machine. The cart shall be designed so that a fully loaded cart can be transported by one person. The cart shall have adequate strength and size to store and transport the dimpling machine, tools, and accessories without damage to the contents or the cart.

3.2.7 Hoses and power cables. Unless otherwise specified (see 7.2(j)), the dimpling machine shall have 12 feet of air hose and power cable connecting the dimpling gun with the control unit. Unless otherwise specified (see 7.2(k)), the dimpling machine shall have 20 feet of air hose and power cable for supplying input air and electrical power to the control unit. The air hoses and power cables shall be oil and abrasion resistant to withstand rough handling in a fabrication shop environment without damage.

3.2.8 Pneumatic system. The pneumatic system shall operate from an air supply pressure of 90 to 130 pounds per square inch. The control unit shall include a pressure regulator, pressure reducer, solenoid valves, check valves, and piping necessary for the operation of the machine.

3.2.9 Electrical system. Unless otherwise specified (see 7.2(l)), the electrical system shall conform to the National Fire Protection Association (NFPA) 79, "Electrical Standards for Industrial Machinery". Unless otherwise specified (see 7.2(m)), the machine shall operate from a 115-volt, single-phase, 60-hertz circuit.

3.2.10 Optional accessories. Optional accessories shall be furnished as specified (see 7.2(n)).

3.3 Performance. The dimpling machine shall be capable of forming dimples in the materials and thicknesses shown in tables I and II for the applicable types and sizes of fasteners. The die face temperature shall rise from an ambient temperature to the die operating temperature in less than 10 minutes. The dimpling machine shall maintain the die operating temperature during either continuous or intermittent operation. The workpiece stretch and distortion and the dimple contour, geometry, and visual inspections shall meet the requirements of 5.3.1 through 5.3.3.4.

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3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(o)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910, "Occupational Safety and Health Standards for General Industry".

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(p)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(q)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(r)), the nameplate shall contain the following information:

a. Nomenclature
b. Manufacturer's name
c. Serial number
d. Machine model designation
e. Power input (volts, total amperes, phase, frequency)
f. Short-circuit/over-current rating
g. Contract number or order number
h. National stock number
i. Date of manufacture

3.8 Lubrication. As recommended and provided by the manufacturer, the dimpling gun shall have oil holes or grease fittings for lubrication of moving parts.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and

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quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The dimpling machine shall be inspected to determine compliance with the requirements specified herein.

5.3 Testing. Unless otherwise specified (see 7.2(s)), the dimpling machine shall be tested in accordance with sections 5.3.1 through 5.3.3.4.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(t)), the manufacturer shall be responsible for supplying all tooling and materials and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(u)). Unless otherwise specified (see 7.2(v)), all tests shall be performed indoors with ambient conditions of 41 to 104 degrees Fahrenheit and 20- to 95-percent relative humidity.

5.3.2 Operational test. The dimpling machine shall be operated in accordance with the manufacturer's standard operating procedures. The dimpling machine shall be operated without forming dimples for at least 30 minutes. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The workpiece shall be a 3- by 12-inch flat sheet of 7075 aluminum alloy, tempered T6 per the American Society for Testing and Materials (ASTM) B209, "Aluminum and Aluminum-Alloy Sheet and Plate", of nominal 0.051-inch thickness. A straight line shall be scribed along one long side of the workpiece 3/8 inch from the edge. The scribed line shall be center punched for 14 uniformly spaced dimples located symmetrically about the 3-inch axis. Pilot holes of the correct diameter for forming dimples for #10-32 UNF screws conforming to National Aerospace Standard Military (NASM) 24694, "Screw, Machine, Flat Countersunk Head, 100° Structural, Cross Recessed, UNC-3A and UNF-3A", shall be drilled. The dies for forming dimples for #10-32 UNF screws shall be installed and aligned. The dimpling machine controls shall be adjusted in accordance with the manufacturer's instructions in the operation manual. Prior to forming the dimples, the length of the test workpiece along the scribed centerline shall be measured and recorded for comparison with the same dimension after the test. The method of making these measurements shall be at the option of the manufacturer provided that the measurement is to an accuracy of ±0.001 inch.

5.3.3.1 Stretch test. A dimple shall be formed for a #10-32 UNF screw in each of the 14 holes in the workpiece. After all dimples have been formed, the length of the workpiece shall be measured along the scribed centerline. This measurement shall be compared with the same measurement taken before the test. The stretch of the workpiece shall not exceed 0.020 inch.

5.3.3.2 Distortion test. The stretch test workpiece shall be measured for convexity and concavity. The convexity or concavity of the workpiece measured across the 12-inch span shall not exceed 0.015 inch.

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5.3.3.3 Contour and geometry. The dimples shall be inspected for contour and geometry in accordance with accepted industry practice. The contour and geometry of the dimples shall be such that the #10-32 UNF fastener conical head properly seats in the dimple and the top surface is flush within ±0.005 inch of the surrounding metal.

5.3.3.4 Cracks and other visual defects. The dimples shall be examined under a 30-power magnifier for evidence of cracks, fractures, or other visual defects. Visible evidence of defects in any of the dimples shall be cause for rejection.

5.4 Optional test. When specified (see 7.2(w)), the following optional test shall be performed. A dimple shall be formed for the smallest and largest size of each type of fastener and for both the minimum and maximum thicknesses of each material shown in table I or II, as applicable to the type of dimpling machine specified. Each dimple shall conform to the requirements of sections 5.3.3.3 and 5.4.

5.5 Acceptance. Unless otherwise specified (see 7.2(x)), operational and performance acceptance tests shall be conducted at the manufacturer's site. Failure of the machine to meet the requirements stated in 5.3.1 through 5.3.3.4 shall be cause for rejection.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the machine shall be preserved, packed, and marked as specified in the acquisition order (see 7.2(y)).

7. NOTES

7.1 Sources of documents.

7.1.1 Government documents. Copies of the FAR and the military specification may be obtained from the U.S. Government Printing Office, Superintendent of Documents,
Mail Stop: SSOP, Washington, DC 20402-9328. Electronic copies of the military specification may be obtained from http://astimage.daps.dla.mil/quicksearch/.

7.1.2 ASTM standards. Copies of ASTM standards may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshocken, PA 19428-2959.

7.1.3 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

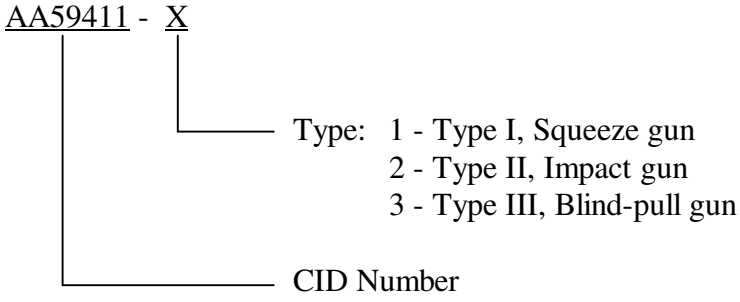
7.1.4 OSHA standards. Copies of OSHA standards may be obtained from the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

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7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document
b. Type required (see 2)
c. Type I gun throat depth, as specified (see 3.2.1.1)
d. Dwell time, if different (see 3.2.2.2)
e. Die sets, as specified (see 3.2.3)
f. Die holder metal thickness adjustment, as specified (see 3.2.4)
g. Die adjustment tools, if required (see 3.2.4)
h. Die heater installation and removal tools, if required (see 3.2.5)
i. Equipment cart, if different (see 3.2.6)
j. Connecting air hose and power cable, if different (see 3.2.7)
k. Supply air hose and power cable, if different (see 3.2.7)
l. Electrical system industrial standard, if different (see 3.2.9)
m. Electrical system voltage, if different (see 3.2.9)
n. Optional accessories, as specified (see 3.2.10)
o. Machine dimensions and weight, as specified (see 3.4)
p. System of units required (see 3.6)
q. Dual system of units, if required (see 3.6)
r. Nameplate, if different (see 3.7)
s. Testing, if different (see 5.3)
t. Test responsibility, if different (see 5.3.1)
u. Certificate of calibration, as specified (see 5.3.1)
v. Test environmental conditions, if different (see 5.3.1)
w. Optional test, if required (see 5.4)
x. Acceptance, if different (see 5.5)
y. Packaging requirements, if different (see 6.1)

7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



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MILITARY INTERESTS:

Custodians:
Navy - SH
Air Force - 99

Reviewers:
Navy - AS, MC, OS
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

(Project 3448-0016)

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